ROS 2 installation and simulator quick start

Developer guide V0.1 | Source software v1.1.0 | 2026-10-11

Luchs A and B software setup. Begin without a vehicle; vehicle commissioning requires the matching hardware and firmware.

Status: Software reference

Choose the software environment

The public v1.1.0 package contains a shared C++ SDK, ROS 2 driver, messages, launch files and robot descriptions. ROS 1 is available as a separate existing download. This guide uses ROS 2.

Environment	Source-supported combination
ROS 2 Jazzy	Ubuntu 24.04
ROS 2 Humble	Ubuntu 22.04
ROS 1 Noetic	Separate ROS 1 package; commands below are for ROS 2

Use a Linux system with the chosen ROS distribution installed and rosdep initialised. Open Developer resources > SDK / API and download the existing ROS 2 v1.1.0 ZIP. No new software archive is supplied by this guide.

Build from the supplied ZIP

1. Create a workspace with a src directory.

2. Extract the ZIP so its top-level luchs_ros2 directory is directly inside ~/luchs_ws/src/. Do not add an extra directory layer.

3. Source the matching ROS distribution, install the declared dependencies and build the workspace.

4. Source install/setup.bash in each terminal used for the example.

```
source /opt/ros/jazzy/setup.bash
# Use /opt/ros/humble/setup.bash on Humble.
mkdir -p ~/luchs_ws/src
# Extract the downloaded ZIP into ~/luchs_ws/src/ here.
cd ~/luchs_ws
rosdep install --from-paths src --ignore-src -y
colcon build --symlink-install
source install/setup.bash
```

A successful build should make luchs_bringup and luchs_driver visible to ROS. If a package is not found, check the extraction layout and whether this terminal sourced install/setup.bash.

Start a simulator first

Choose one launch command. dry_run:=true selects the built-in software simulator instead of CAN; no physical chassis is needed.

```
ros2 launch luchs_bringup luchs_a.launch.py build:=4ws dry_run:=true
# Or use the Luchs B draft-LCI simulator:
ros2 launch luchs_bringup luchs_b.launch.py dry_run:=true
```

For another A build, replace 4ws with df, fa or da. In a second sourced terminal, inspect the node and reported state:

```
ros2 node list
ros2 topic echo --once /luchs_driver/state
ros2 topic echo --once /diagnostics
```

Scope note. A successful simulator run confirms software setup only. It does not establish that LCI, IMU feedback, payload switching or other planned functions exist on a delivered B chassis. These examples were checked against source files, not executed on a vehicle for this release.

Before connecting a vehicle

- Match the A build or B interface to the delivered unit.

- Use the CAN connection guide to confirm the adapter, harness and bitrate.

- For B, the default launch selects draft LCI. Select legacy_b only for a unit confirmed to use VCU V2.6; neither choice is a firmware update.

- Use the approved operating manual for control selection and emergency-stop checks. Complete commissioning before issuing motion commands.

Source and edition

Prepared and reorganised by RJ Tech from the already-public Luchs ROS 2 v1.1.0 package by TerraAgent Robotics. Source licence: Apache-2.0. Source text and examples were shortened, explained and scoped for this topic. Original archives and protocol files remain unchanged.

Source file: README.md

Source file: luchs_bringup/launch/luchs_a.launch.py

Source file: luchs_bringup/launch/luchs_b.launch.py

Original public software archive | Source licence